

ActInf GuestStream #024.1 ~ Stephen Grossberg, "Explainable and Reliable AI"

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DANIEL FRIEDMAN: Hello and welcome, everyone! It is June 28th, 2022 and we are here in ActInfLab GuestStream number 24.1. Today we're here with Professor Stephen Grossberg and the agenda will be as follows. First, Ali will provide a short introduction. We will then play a 45 minute prerecorded video followed by a Q&A. So thanks, everyone, for joining. And Professor Grossberg, really appreciate joining. And I'll pass to Ali for the introduction. ALI RAHMJOO: Hello and welcome! I'm Ali. I'm an independent researcher from Iran. I'm very happy and excited to be here and be able to speak with {Professor and} Professor Grossberg today. So I'd like to thank Professor Grossberg for joining us. Stephen Grossberg is the Wang Professor of Cognitive and Neural Systems and a professor emeritus of mathematics in the statistics, psychological and brain sciences and biomedical engineering at Boston University. For more than 50 years, he has led pioneering research in discovering and developing neural design principles for autonomous adaptive intelligence based on biological and machine learning. His neural network models have been applied to many large scale problems in engineering and technology, including the design of increasingly autonomous adaptive algorithms and mobile agents. In fact, this is what Karl Friston says about him. Whenever you claim to be the first to do this or that in artificial intelligence, it is customary and correct to add, with the exception of Stephen Grossberg. Quite simply, Stephen is a living giant and foundational architect of the field. Professor Grossberg is the recipient of the 2015 Norman Anderson Lifetime Achievement Award of the Society of Experimental Psychologists, the 2017 Frank Rosenblatt Award of the Tripoli Computational Intelligence Society and the 2019 Donald Wahab Award of the International Neural Network Society. His latest book, "Conscious Mind, Resonant Brain," as a culmination of his decades long research written in a rather non-technical and conversational style, is published in 2021 by Oxford University Press and is the winner of the Association of American Publishers in 2022 Prose Award for the Best Book of the Year in Neuroscience. Now I'll pass it to Professor Grossberg and then we'll

continue with the 45 minute prerecorded lecture. FRIEDMAN: If you'd like to say hi... otherwise, I'll begin the video. GROSSBERG: (I just saw my face frozen on the screen!) Well, I'm delighted to be here, and I hope you find some points of interest in the lecture. And I'll look forward to the Q&A. Ali has prepared a series of questions that I've thought about and have some prepared, sketched answers. And then after that, if you're still interested, I'm happy to do. Live Q&A about anything related to the topics of the day. FRIEDMAN: OK! - On to the main course. I will play the video now. (And you won't hear anything on the live stream. I'll crop it, and the audio will be coming through fine now.) GROSSBERG (via video): Hello! I'm delighted to be able to speak to you today about a topic concerning artificial intelligence, which, as you know, is very much in the news these days. And I'll be contrasting two very different approaches to artificial intelligence. But to do that, I need to pull up my PowerPoint slide cards and share them with you and let me maximize them and minimize my space. So my topic today is explainable and reliable. A.I. Comparing deep learning with adaptive resonance. This lecture is based on the following article from this year, which is both open access and on my web page. The article summarizes core problems of deep learning, such as its untrustworthiness. Because it's unexplainable and it's unreliable because it experiences catastrophic forgetting. The article explains how adaptive resonance overcomes these problems and indeed overcomes 17 problems of deep learning and outlines a blueprint for achieving autonomous adaptive intel diligence. The article is part of a frontiers in robotics special issue about explainable A.I., whose editors wrote and I quote, though deep learning is the main pillar of current A.I. techniques and is ubiquitous in basic science in real world applications. It is also flagged by AI researchers for its black box problem. It is easy to fool us and it also cannot explain how it makes a prediction or decision. In other words, deep learning is not trustworthy. No life or death decision, such as a medical or financial decision, can confidently be made based upon a deep learning prediction. Deep learning is in the back propagation algorithm for learning how to predict out the factors in response to win protect is that propagation was based on perceptron learning principles that Frank Rosenblatt's started to introduce in the 1950s. It has a complicated history, which Yorke and Schmidt Huber beautifully refuted in an article from this year. Major contributors include Shinichi, Amari for Bros and David Parker. Perhaps one would say that it reached its modern form with simulated applications in Paul's 1974 paper before being popularized 12 years later by Rummel, Hart, Hinton and Williams is a schematic of a back propagation circuit with printed word from a survey article by Gail Carpenter of Neural Network Models. Any information flows,

feedforward from an input stage to an output stage. Learning is supervised by an external teacher who on each trial defines a target or desired output. The teaching signal is the error or mismatch between the actual and the target outputs. The teaching signal in level three of adaptive weights and level left to have no network pathway whereby to reach from F3 to EC2 within the algorithm. So the algorithm uses an artifice called way transport, which physically lifts the weights from here and moves them there so that they can be used to control learning. Well, this is clearly a non-local operation as well as being clearly non-biological. Back propagation learns through slow learning, which means that the adaptive weights change just a little to reduce error on each learning trial. That requires many trials, that is to say, many repetitions of the whole database to learn possibly hundreds or thousands of trials. This is to be contrasted with fast learning where adaptive weights zero error signals on each trial, just as we can learn a phase that we see just once and remember it for a long time. If back prop try to use fast learning would become wildly unstable. Catastrophic forgetting also occurs in back prop so that during any learning trial, an untreated active part of its learning memory can unexpectedly collapse. So deep learning, which is based on back propagation, is thus neither reliable nor trustworthy. But why is this? One reason is that all inputs a process by a shared set of learned weights, the algorithm cannot selectively buffer learned weights that are still predictably useful. In particular, there's no attention mechanism. This problem occurs in any learning algorithm whose shared weight updates follow the gradient of the error in response to the current batch of data points while ignoring past actions. There have been multiple efforts to fix back propagation. One is to selectively slow learning on the weights, important for learning by optimizing parameters using the Bayes rule, as Kirkpatrick Atoll suggested a few years ago. But that assumes an omniscient observer who can discover and alter the important weights as well as non-local computation such as the Bayesian computation. The same problem occurs with evolutionary algorithms and diffusion based neuromodulation and other approaches to try to fix spectra from these efforts to overcome catastrophic forgetting created additional conceptual and computational problems. I view them as adding epicycles to ameliorate a fundamental flaw in the model, which to me is reminiscent of adding Epicycles to correct problems in the Ptolemaic model of the solar system. As we all know, the Copernican model that we now accept didn't require epicycles. Perhaps this is why Geoffrey Hinton, who played a key role in developing both back prop and deep learning, said in an Axios interview a few years ago that, quote, he's deeply suspicious of back propagation. I don't think it's how the brain works. We clearly don't need all the labeled data. My view is throw it all away and start

over. I would claim we don't have to start over because these problems were solved in the 1970s and 1980s in particular in the first issue of the journal Neural Networks in 1988. I had an article that listed 17 Problems of Back Propagation that are overcome by adaptive resonance. And here they are with regard to not needing to label data. I noted in the third item here that self-organized, unsupervised, unsupervised learning frees us from needing labels all the time as to slow learning. I noted that in art you can have fast or slow learning. Indeed. Or can learn to classify an entire database using fast learning on a single learning trial, says Gale Carpenter. And I showed in the 1980s. Rover Auto overcomes all 17 problems of back propagation without ex bicycles. Furthermore, all the core AI predictions have been supported by subsequent psychological and neurobiological data. Indeed, art is the principle, biological and technological theory. Unlike back prop and deep learning, which are just AI algorithms, Art has explained data from hundreds of experiments, and it's made scores of predictions that have subsequently received experimental support. But why has it been so successful? There are a number of reasons, but one of them is that art can be derived from its thought experiment about a universal problem an error correction that I published 40 years ago in Psychological Review. The thought experiment asks the question, How can a coding error be corrected if no individual cell knows that one has occurred? Let me quote from my paper The importance of this issue becomes clear when we realize that erroneous cues can accidentally be incorporated into a code when our interactions with the environment is both and will only become evident when our environmental expectations become more demanding. And even if our code perfectly matched a given environment, we would certainly make errors. As the environment itself fluctuates. So I was talking about autonomous local learning in a changing world, a purely logical inquiry into error correction is translated at every step of the thought experiment into processes. Learning autonomously in real time with only locally computed quantities. Moreover, the thought experiment uses familiar environmental facts about how we learn as its hypotheses and short circuits naturally emerge where these facts are familiar because they are you. They are ubiquitous environmental constraints on the evolution of our brains. And since we're living with them all the time, they become familiar because of this. Universally, earth circuits made us in some form be embodied in all future truly autonomous. Adaptive, intelligent devices with a biological or artificial, or it has probably for this reason already been used in many large scale engineering and technological applications. almost immediately after it was introduced, it began being used because it succeeded in benchmark studies against machine learning, back propagation, statistical methods, genetic algorithms either getting

much better accuracy or much faster training, speed or both. It's also used in applications where other algorithms totally fail, such as the Boeing Company's part design, reuse and inventory compression application. That's just one of many large scale applications in engineering and technology, some of which can be found on our Tech Live Web page at B.U. dot edu. The Boeing parts design retrieval system in particular was used to help design the Boeing 777. And to do that, you needed fast learning and stable memory to learn and search a huge and continually growing non stationary parts inventory. At the time of this application. There are already 16 million, 1 million dimensional vectors that were used to describe each of the parts and you have to be able to quickly search the inventory if you want to find a part to use in a new plane, especially if your new design might have a part in the inventory that was similar to it. Finding it and slightly modifying design could save millions of dollars in fabrication costs. Satellite Remote sensing is another large scale application that art was used for very soon, and Gail Carpenter and a colleagues took the lead here for example, using a very small number of pixels of ground truth of 17 vegetation classes. They used art to automatically complete these maps using remote sensing data. Or did it in a day rapidly and automatically, it gave a confidence map for each pixel and the pixels with 30 meters in scale, which was small enough to see roads. This contrasted with an AI expert system which took a whole year to do with and it had to derive at ad hoc rules from experts. You had to correct up upwards of a quarter of a million site labels. And even so, the pixel size was an order of magnitude larger. Gail went on with her colleagues to study information, fusion and remote sensing. Let's say you have multiple observers. Each of them may be using different labels. The labels may also be incomplete or missing or even incorrect. And the task was to derive consistent knowledge from potentially inconsistent data to automatically learn and stably store one too many mappings. And along the way, Gail and a colleague showed how to self-organize a hierarchy of cognitive rules, including confidence measures between these different levels of the hierarchy. There's been continual work on art. Some more recent work was summarized in a special issue of neural networks just in in December 2019 that was edited by Donald Bunch, who started the special issue with a general overview of neural network models that I and my colleagues developed, and then went on in a long and detailed article with several collaborators to provide a survey of adaptive resonance theory neural network models for engineering applications to the present time, so that propagation and deep learning are a feedforward adaptive filter. But art is more than that. In fact, art is explainable self-organizing production system in a non stationary world. What do these words mean? Or self-organizing.

Because they can autonomously carry out arbitrary combinations of unsupervised or supervised learning trials with the world as its only teacher. It's a production system because it uses hypothesis testing to discover and learn rules by a top down matching process that focuses attention on critical feature patterns. These are the patterns that predict behavioral success while suppressing irrelevant features, or it's explainable using both its activities or short term memory. As STM traces and its adaptive way to a long term memory, LTM traces, activation dynamics, learning dynamics, observing the STM traces in a critical feature pattern. Explain in what recognition categories will learn to code and what features predict goal oriented actions and particularly the long term memory traces in the fuzzy art map algorithm translate into explicit fuzzy if then rules that code. What combat actions of critical features in what numerical ranges effectively control predictions thereby illustrating one of many examples where neural networks can learn rule based behaviors. Art includes a bottom up adaptive filter, a feedforward neural network, as I've observed already, but that's supplemented by top down learned expectations and two types of recurrent inhibitory feedback interactions that help to choose the recognition categories and the critical features. Notably, top down expectations use what Joe Carpenter and I call the art matching rule to learn how to focus attention on critical features that control predictive success. The art matching rule is another way of talking computationally about the process of object attention how we pay attention to salient objects in the world, and we show how it stabilizes learning and thereby avoids catastrophic forgetting. Remarkably, and this has been supported by many data, the art matching role can be realized by a top down modulatory on center or surround network. Where? What does this mean? Well, let's say we have bottom up inputs from external features to feature selective cells that get stored in short term memory. Let's say we activate a recognition category which has previously been learned and tries to read out its learned excitatory prototype. But it can't fully do so because it also reads out an inhibitory of surround that's broader than the prototype. So this is approximately one excitatory against once inhibitory. It can only give you one modulatory on center. But if you have both bottom up inputs and the top down expectation simultaneously active, then within the bounds of the prototype. If you also have a bottom up feature, you have two excitatory against one inhibitory and those features can be selected, gained, amplified and synchronized. To start focusing attention on this critical feature pattern. Well, outlier features the ones that aren't within the prototype only have one excitation against one inhibition or suppressed. And in 1999, I was able to begin to understand how lamina cortical circuits carry out object attention. In particular, layer six of a higher cortical area can

activate layer six of a lower cortical area, either directly or via a five, and then it can fold up the layer for to modulate in, on center and to inhibit and of surround. So attention x by a top down modulatory on center of surround network by a folded feedback within lamina neocortex. And this is one example of the paradigm of lamina computing that I introduced, which ask Why are all neocortical circuits organized in layers? And how do laminar circuits give rise to all kinds of biological intelligence? Adaptive resonance sets is the story because attended feature clusters react elevate their bottom up pathways, activated categories reactivate their top down pathways closing in excitatory feedback loop between features and categories, giving rise to a feature category resonance that synchronizes, amplifies and prolongs system response between the attended critical features and the category to which they are bound. And it's this resonance that triggers fast learning in the bottom up and top down. Adaptive waits, which is why I have called this theory adaptive resonance theory. Moreover, I've done a lot of work since then, showing that all conscious states or resonant states and these feature category resonances are one example of that, one that supports conscious recognition of visual objects and scenes. There's a lot of data support for our predictions. It's well known that attention does happen on center of surround a circuit behind it, and that attention can facilitate matched bottom up signals. Many other data as well. So now we can say more about why art is explainable or trustworthy in short term memory. It's because the critical feature patterns determine the attentional focus that controls information processing. And you can just read off what those features are in long term memory. Again, it's the critical feature patterns that determine the adaptive way to learn by the bottom up adaptive filter and the top down learned expectation. So, you know also what these weights are encoding less reliable and avoids catastrophic forgetting because outlier features that are not in the critical feature pattern is suppressed so that only the predictive features are processed and coded. Well, it's a production system because it carries out a kind of hypothesis testing. And this is nicely illustrated in the simplest art model called or one that Gale Comaparator and I published in 1987 or at one has an attentional system that does all the category learning and the expectation learning and the paying attention that interacts with an orienting system which is activated when there are big enough matches in the attentional system and thereby drives a reset and search for novel. Look at a match in categories. Here's a schematic of the yard hypothesis testing and learning cycle. So let's say you have a bottom up feature pattern coming in. There may be many, many active bottom of features, but I'll draw just one arrow here for simplicity. But that vector of input features can activate a distributed pattern of feature detector cells. Some may be very active,

some not so active, some not active at all. And as this is happening, each of these active pathways is trying to turn on the orienting system. So there might be quite a few inputs converging here, but as the features are activated, each of them tries to inhibit the orienting system and there is many features as there are inputs. So this excitation and addition of balance, keeping the orienting system quiet as the feature padding goes, the adaptive filter and chooses a category. That category reads out of learned top down expectation that obeys the art matching rule, which can suppress a mismatch features, thereby reducing the amount of inhibition on the orienting system and raising the question when you have too little inhibition and too much excitation. How big a mismatch will activate the orienting system and cause reset and that ratio is determined by what's called vigilance, which I'll say more about soon. But if you don't have enough inhibition, then the orienting system gets activated. It equally activates all the cells in the category layer because it doesn't know which cell may be active or not. So it causes and novelty sensitive, nonspecific bursts of arousal, novel events or browsing, thereby selectively shutting off the active category, eliminating its top down expectation, and unmasking the original feature added, which can again go through the adaptive filter. However, now this previously this confirmed category remains off and the category level is we normalize sort response to the same input bad and with the new category and you go through this cycle of resonance and reset until you get a good enough match to either learn a new category or select a previously learned category. And it's a theorem that as categories are learned through this matching process, search automatically disengages leading to direct access without search to the globally best matching category. Explaining, for example, how we can quickly recognize familiar objects like your mother's face. Even if, as we get older, we store enormous numbers of additional memories so you don't have to search your whole repertoire. When you see Mom, you get direct access and quickly say, Hi, Mom. There's a lot of support for the hypothesis testing cycle. One source of support is from event related potentials, also called human scalp potentials, which shows correlated sequences of three different evoked potentials during oddball learning tasks. An experiment that John Paul Bank and I reported in the eighties where you'll get a p120 for a mismatch and then 200 for the arousal activated by the ordering system and a p300 for the short term memory reset of the category layer, thereby supporting the processing stages of the search cycle. There was also physiological data from a temporal cortex where categories are learned early on from the lab of Bob Desdemona, who showed an active matching process that's reset between trials during this kind of event. There's also classical data about hippocampal mismatch dynamics. It's

known that novelty potentials subside as learning proceeds from numerous experiments. This is as the orienting system is disengaged, and there's more recent data using multiple electrode studies from the lab of Cheryl Miller, from pre-frontal cortex and simultaneous recordings in active campus. And they show this rapid object associative learning may occur in prefrontal cortex, which is a projection of inferred temporal cortex, one of the stages of category learning. While the hippocampus may guide neocortical plasticity four by signaling success or failure. Well, this is just what happens when the attentional system interacts with the orienting system. There's also complimentary computing in art. In particular, the attentional and orienting system was a complementary as manifested by the fact that two event related potentials are complementary processing negativity and n200 processing negativity is activated when there's a top down match in the attentional system. The N200, as I just noted, is activated when there's a mismatch that activates the orienting system. And you can just look across these four rows and see that these two kinds of ERP potentials are manifestly complementary. As illustrated, the complementarity of the attention and orienting systems. So this leads us to discuss another paradigm introduced, which I call complementary computing, that has what is the nature of brain specializations? Complementary computing introduces new principles of uncertainty and complementarity that clarify why there are multiple parallel processing streams with multiple processing stages in our brains. And a beautiful example of that is this famous image of the macro circuit of the visual system from David Ranson and his colleagues, where you can see these multiple parallel processing streams and the multiple stages needed to achieve, I call hierarchical resolution of uncertainty. So what our complement complementary properties, their analogies like a key fits into a lock for puzzle pieces fitting together in words computing one set of properties that a stage prevents that stage from computing a complementary set of properties. These complementary parallel processing streams a balance against one another. It's a very yin yang kind of situation, and interactions between the streams overcome their complementary weaknesses. In fact, there are many complementary processes that are known in the brain that have been modeled. Here is just a of them. There are many more. So this is a basic principle of brain organization. So in summary, so far back propagation and deep learning do not have thought to memory activation patterns, including critical feature patterns, so they can't pay attention. Indeed, they don't have any fast information processing, nor do they have long term memory at the moment. Expectation So they can carry out hypothesis testing using interaction short term and long term memory traces. Indeed, there's no neural architecture. There's just an algorithm in this really great

contrast with complementary computing, which discusses the global organization of brains. From the very start, it was shown how easy it is to get catastrophic And Carpenter and I showed it in art when we would shut down the matching rule. Then we demonstrated you could get catastrophic forgetting if you had just four input vectors A, B, C, D presented in the order ABCD, ABCD, and so on. If they obeyed very simple subset relationships and here's a computer simulation of that. Here you don't have the matching rule is a, b, c, d, a, b, c, a, D and you see A is coded by category one here by category two here by category one. Year two we are never settled down. But as soon as you impose the matching rule, learning is complete by the second trial and after that point you get direct access Well, let's say a little more about vigilance. Vigilance determines what features are learned in the critical feature pattern. It clarifies our brains, learn concrete knowledge for some tasks and abstract knowledge for others. So in particular, high vigilance leads to learning of narrow, concrete categories, like a category that fires selectively to a frontal view of your mother's face. Low vigilance leads to learning of broad and abstract categories, like everyone has its face. It should be emphasized that critical feature patterns are explainable at every level of vigilance. It's known from physiological experiments by does moment again that there's vigilance control in the infra temple cortex, which they showed by studying easy versus difficult discriminations in monkeys and the difficult condition which you'd assume would give you their children's. As expected, you had enhancement of the responses and sharpen selectivity to the intended stimulate. How is vigilance computed? Well, let's say of input vector it in states of vector of activities, feature detectors at the same time as it tries to activate the warning system. But it this does so multiplied by a parameter ρ , which is a sensitivity against gravity that vigilance. And as these features get in state, they try to shut off the orienting system. And if the excitation is less than the inhibition, the orienting system stays quiet. So the system can resonate and learn. But if inhibition isn't strong enough, the orienting system gets activated. You get reset and search for new categories. This is a very simple computation because you have an orienting system that's complementary to the attentional system. Well, how do you change vigilance based on predictive success? For this, we have to go from unsupervised to supervised start models. So we'll have an unsupervised art, a model in unsupervised, or B model linked together by a learned associate of map as occurs in the art map. And the key point is you can have an input here that can create an output there because you have both bottom up and top down in actions at all of these levels. So in this way, you can learn many to one and one too many maps. One example of a many to one map is let's say you're trying to

categorize visually processed a letter A, which comes in multiple fonts. You'll learn various visual categories. Are they based on visual similarity? At the same time, you're learning auditory categories for saying A and then the associated map all of these visual categories of different age to saying, Hey, but it could have been here that these inputs were symptoms, test treatments in a medical database prediction example and you're predicting length of stay in the hospital. The possibilities here are endless. And then in many applications or let's say you're trying to figure out what this images and you've learned to say that's a dog. But today you say it's rover and that causes a major mismatch, which drives the search to focus attention on the particular combination of features in this talk that will identify it as Rover. That leads to learning of a visual category of Rover in auditory categories of the name, rover and associated map between them. And you can now simultaneously store expert knowledge about that image. Well, how do you conjointly minimize predictive error and maximize generalization so that you minimize using memory resources? Let me read you an answer and then show what it means in images. Match tracking realizes in kinematics a learning principle, namely, given a predictive error, vigilance increases just enough to trigger search and the sacrifices, the minimum generalization to correct the error. So let's say you've made a prediction that must mean that vigilance is less than the analog between bottom up and top down. But let's say now you have a mismatch. Well, that'll lead to a match tracking signal that vigilance up till it's just above the analog match, just big enough to drive the search. So you've given up the minimum amount of general deception to correct the error? Well, all lot mechanisms like vigilance, control realized in lamina cortical insular mixture because the answer is yes. My Ph.D. student Extra Saatchi and I showed this by developing the synchronous matching order, a smart model which introduce a lot more neurophysiological and anatomical verisimilitude into the model, including spiking dynamics, lamina cortical circuits, interacting with specific and nonspecific galactic nuclei. This is another example of what I'm going to computing, and here's a schematic of the model. You see all the cortical layers with identified cells, a hierarchy of cortical regions interacting with specific thalamic nuclei in nonspecific thalamic nuclei, a ton anatomical data got functionally explained in this way and many other data as well. For example, we showed if you have a good enough match between bottom up and top down, you're going to get fast gamma oscillations during a tension. There was quite a bit of data about that already, but we also showed if you have big enough mismatch, you'll get slower data oscillations. That wasn't well known, but since that time there have been experiment in at least four labs in three different parts of the brain confirming that

prediction. Most important, vigilance control was shown how to be a control by mismatch mediate the colon release so big enough mismatch in the nonspecific thalamic nucleus activates nucleus based cells. The miner that releases acetylcholine in lay off high shields across the cortex, reducing after hyper polarization currents and causing vigilance to go up. And I also showed that breakdown in acetylcholine modulation can help to explain the symptoms of multiple mental disorders so as to memory consolidation. We know there's a dynamic phase of memory consolidation, while the input exemplar still drives memory search and before direct access occurs. But what if the orienting systems cut out? What if you have a lesion in the hippocampus? Well, then, as occurs in medial temporal amnesia, you get unlimited anterograde amnesia because you can search for new categories. You get limited retrograde amnesia because you could have direct access to previously learned categories. This is a failure of consolidation, which is mediated by the ordinances. So you get defective novelty reactions because that is also mediated by the orienting system and memory consolidation, novelty detection mediated by the same structures for the same reason. It's normal priming because priming occurs within the attentional system. Learning of the first item dominates. You can get some learning, but you can't then search. And there's an impaired ability to attend to relevant dimensions of stimuli again because you can't search. So now where does intra temple cortex fit in the larger brain? I introduced the predictive or taught algorithm model in order to show how the prefrontal cortex, among other things, learns to control all higher order intelligence. You can find that in a 2018 paper on my web page, results were published in Open Access and in this macro circuit, these green areas of prefrontal cortex control processes like working memory, learning plans, prediction and optimized action. These regions in red control processes like reinforcement learning, motion motivation, adaptively timed learning. The category learning I've talked about in I.T is just in those two regions. All these processes control visual perception and there are detailed models of all of these regions and their interactions. Now in each brain region, in nature and in predictive art carries out a different function, contrasting really dramatically with the whole mad genius organization of the typical deep learning network. So I told you just a little bit about some aspects of cognition and why they're explainable, but if you put in all the biological models of perceptual cognition, emotion and action, they're all explainable. And then you can observe how perceptual and cognitive processes work, like excitatory matching, match based learning to create self stabilizing, attentive and conscious representations of objects and events that embody increasing expertise about the world. Moreover, complementary spatial and motor processes that I couldn't mention

at all use inhibitor free matching and mismatch based learning to continually update spatial and motor representations to compensate for bodily changes throughout life. Taken together, they provide a self stabilizing, perceptual and cognitive front end for conscious awareness and knowledge acquisition, which can intelligently manipulate the more labeled spatial and motor processes that enable changing bodies to act effectively on a changing world. And when you put them all together, they provide a blueprint for designing autonomy. This adaptive algorithms and mobile robots with behaviors humans can understand and control because they're both explainable and reliable. See my web page, [sites \(dot\) BU \(dot\) EDU \(slash\) SteveG](http://sites(dot)BU(dot)EDU/slash/SteveG), for these models. And with that, I'd like to thank you very much for your attention.

[End of video] GROSSBERG (live): Before we get started, what I want to say so I should say that everything I talk about and much more is in my book *Conscious Mind, Resonant Brain How each Brain Makes and Why*. For those who don't know, it's self-contained and non-technical. It's written in a conversational style so that people will know nothing about the mind or the brain and enjoy reading it. And I have friends who are a rabbi, a minister, a painter, a gallery owner, a lawyer, a social worker who've all been enjoying reading it all. So it's a big book. It's almost 800 double column pages with over 600 color figures. So everything is illustrated, but instead of costing hundred \$50, it costs \$35 for the hard copy and only \$17 for the Kindle. Because I spent a lot of my own money so that people who are interested in the book and read it. And one other comment, if people do have questions or comments about my lecture or anything, they're reading the book. My email is just [Steve STV The EU Boston University](mailto:Steve.STV@TheEU.BostonUniversity.edu) study the EU and I'll be happy to try to apply, so thank you. Help some researchers and explainable AI people like Leonardo, Giovanna and Antonio de Checco demand that any explainable I should at the very least meet these four criteria to be fair, not biased in one way or another, to be accountable or reliable, to be secure against malicious hacker attacks and not to be fooled easily and also to be transparent. Now you explained how adaptive resonance theory or art. And by the way, I got to say, I love your creative and clever use of acronyms for your models. My favorite one is Sovereign Model, self-organizing vision, expectation recognition, emotion, intelligent, goal oriented navigation, if I'm correct. Amazing. Anyway, you explained how art can address and overcome the issues of accountability, security and trends, currency of current deep learning approaches. But it seems that this fairness issue, a.k.a. the problem of algorithmic bias, has also been a growing concern lately, especially since it's regarded by some researchers like Antonio Badia as a practically intractable problem. So I wanted to ask, in what ways do you think art can contribute to the ongoing quest for mitigating

this problem? Well, when a fairly sent me this question, I said, Well, first I'd like you to send me a definition of algorithmic bias that will clarify what you have in mind so that I know what I'm trying to respond to. And you wrote me that you part the term from body is both the information manifold. And you sent me a quote from page 247 that I will quote in part before I respond to that background information. So there are two main reasons for algorithmic approach to decision making that may result in unfair outcomes, either at the individual or group level. One is that data used is biased, and another is that the algorithm analyzes the data in such a way that it yields biased results. The basic point to remember is that algorithms are designed to achieve a certain goal not created naturally by evolution or accident. Thus, most algorithms are written to detect certain patterns of interest for a particular objective, not just any pattern, to be able to pick out some patterns and disregard others, program is build a model of the data by listing expectations about what data should be like in order to qualify as relevant to the problem. Well, as I'll explain below, self-organizing learning classification and prediction models like adaptive resonance Theory or art overcome all the problems. It's a general purpose device. But why don't I try to answer that as part of my replies to at least subsequent questions? Okay. Thank you so much. Now, as you also mentioned in your lecture in 1988, you pointed out 17 issues with back propagation and one of your most famous and highly cited papers on nonlinear neural network. So it's been 40, 34 years now. Now, do you see any fundamental Copernican change of perspective happening and deep learning research or we still keep we're still adding Epicycles upon Epicycles to our Ptolemaic model? Well, you've sort of anticipated what I'm going to say, and as I said in my lecture, there is investigators and I mentioned on Kirkpatrick and Vella as well have recently attempted to modify deep learning to overcome some of those problems. But as I just mentioned by lecture noted that at least in my mind, they're like epicycles that are added to a kind of Ptolemaic model of the solar system to overcome some of its problems. But as we all know, the Ptolemaic model ultimately crashed because it was both qualitatively and quantitative wrong and they could only be solved by throwing out the Ptolemaic model and replacing it with the Copernican model that became the basis for modern astronomy and astrophysics. So art overcomes foundational, deep learning problems that can't be solved using Epicycles, and it's already done. It Shortly after I introduced it in 1976, and it can't be overemphasized. As I noted in my lecture, Deep learning is untrustworthy because it's not explainable because it can experience And that happens for a basic reason. Deep learning, just like back prop, which is its learning engine, is just the sheet for with adaptive filter. So as you note in your question, I

described these two problems in addition of 15 others. In my oft cited article that I published in 1988 and the first issue of neural network, and I also showed that that or that already solved the problem of the 1976. What I find sad is that back propagation and deep learning architects like Jeff Jensen, who knows all of this background, never mention this history and keep talking about making deep Explain the brain. But it can't explain the brain cause its foundation is contradicted by basic psychological and neural data. Yes, some great deep learning community. I like comparative discussion and criticism, but I don't like solipsism and science. Great. Thank you. Now on slide number 50 of your presentation, you pointed out that art is inconsistent with models where top down matches suppressive such as Bayesian explaining away. A similar view is evident on page 195 of conscious mind resonant brain, to which you also adds one of many serious problems of the Bayesian models is that fully suppressive matching circuits cannot solve the stability plasticity dilemma. Would you care to further elaborate elaborate on this point? Sure. My lecture and my book summarizes my book. My lecture couldn't go into a lot of it. Some of the copious psychological anatomical and your physical logical evidence expectations which obey what I call the old matching rule or match against bottom up input patterns. And as the lecture briefly noted, the matching rule is defined by a modulatory on center off surround network, and the modulatory center is excitatory. However, acting by itself. So it can't fully excite its target It can prime them sensitize or modulate them to be ready to fire vigorously when matched, bottom up inputs arrive, and when there is a good enough between the parent input and an active up down expectation, that's reading out a circuit that obviously ought matching role. That's when you get what I noted in my lecture, what I call a feature category resonance, because it develops between the matched or intended features and the recognition category that they activate. that synchronizes and amplifies the matched features while suppressing the mismatch feature. And that sustained resonance is important because it sustained long enough to drive learning in the more slowly varying adaptive weights of the active bottom up filter and learned top down expectation. And because resonance triggers learning that I call the theory adaptive resonance theory and the old matching rule avoids catastrophic forgetting as I briefly mentioned in the lecture, because it suppresses irrelevant features, using it to off surround while it's amplifying and focusing attention on the critical features that regulate both bottom up and top down learning as well as. Successful predictions because they're relevant. They've been selected by previous learning experiences to which discover this set of features that are predictive. Also in a given situation. And along the way, not only does the matching rule achieve causality and predictions,

although as the world changes, you have to update your plausible explanations. It also serves as plasticity dilemma. In brief, purely suppressive match and can't do any of this. It shuts off the expected data and so it can't focus, attention or learn about it. And there is fully suppressive matching and spatial and modal learning, but that isn't learning to be expert about the world. I can explain that more if you want to know, but that's also in my book and these two kinds of learning, the excitatory match based learning and the inhibitory mismatch learning or computation complementary. It's another example of complementary computing and the match based learning goes on in the ventral or what cortical stream and the mismatch learning goes on in the where or dorsal cortical stream. The what stream for perception and categorization of prediction. The Where Stream for the spatial representation in action. And then you need what, where and when, what interactions so that you can reach for and otherwise engage through approach. And if you look at reach for approach to things that you've recognized. Now, following from the previous question and considering that the free energy principle and active inference framework has works in progress are related to predictive coding and Bayesian brain hypothesis. What is your view on the extent of compatibility between art and active inference? Because despite some prima facie similarities between the two, do you see them as fundamentally incompatible or irritants salable? And how could this how could this issue be rigorously evaluated and positively resolved in terms of reconciliation or integration of art and acting and active inference or otherwise? Because you seem to add some more context here, Smith. Smith at all in their recent paper, an active inference approach to modeling structure learning. Have it stated that although there they have not explicitly incorporated arts in a top down attentional and feedback mechanisms, there are mechanisms within their active inference based model which they believe are quite similar to top down and bottom up feedback exchange in art. So there seems to be some degree of disagreement about the compatibility between the two frameworks. Well, let me try to respond to the two parts of your question. So I'm not going to try to talk about Smith at all for a moment. Let's talk about free energy. And I like getting definitions on the table because it's really so frustrating to try to remember what something is when someone's talking about it. So I go to Wikipedia.

Wikipedia writes in part that the free energy principle asserts, quote, "that systems minimize the free energy function of their internal state... which entailed beliefs about, hidden states in their environment. The implicit minimization of free energy is formally related to variational Bayesian methods and which originally introduced by Karl Friston as an explanation for embodied perception in neuroscience, where it's also known as active inference." We all know

he's a brilliant and very insightful man. The free energy principle describes the behavior of a given system by modeling it through a Markov blanket that tries to minimize the difference between their model of the world and their sense and associative perception. This difference can be described as surprise, and it's minimized by continuous correction of the world model of the system. One more part of the quote The free energy principle has been criticized for being very difficult to understand, even for experts. And the mathematical consistency of a theory may have been questioned by recent studies. Discussions of the principle have also been criticized for invoking metaphysical assumptions far removed from testable scientific predictions, making the principle non-falsifiable. And in a 2018 interview, Friston acknowledged that the free energy principle is not properly falsifiable. So that's risk. So so my main concern with the free energy principle, just like any theory about how brain makes a mind, is how much data can it explain in a principle and unifying way? That's what we do in science. We develop theories to imperfect data. And in the case of the free energy principle, from what I can see here, there is essentially no data. And you can correct me if I'm wrong. It therefore cannot be evaluated as a physical theory at all. And there's a basis for this problem. Our brains are designed to autonomously learn in real time to a changing non-stationary world that's filled with unexpected events like today, we're experiencing an unexpected event that I didn't know till that I'd be enjoying your company today. Optimization principles were designed to cope with stationary dynamics. One's rules and probabilities do not change through time, so it's not possible to, quote, minimize the difference between their model of the world and their sense in associative perception, unquote, because there is no predefined model of the world, which is always changing in unexpected ways. So you need a theory about how the world changes. Surprise occurs in art when there's a big enough mismatch between an input and the currently active top-down expectation of a category that it's activating. This mismatch activates the orienting system that I briefly discussed in my book, which interacts with the attentional system where the category learning does occur. And as I illustrated in our discussions of search in this, you learned that it drives hypothesis testing or memory search to discover a better match or to begin to learn new category. So order discuss a better match. In the case where the system was attending, some other familiar features when the new event that occurs but the features are new when put have previously been categorized. That's why they're familiar. And then the orienting system very quickly shifts attention to the matching category. And you resonate on you recognize it consciously, often, or it begins to learn a new category when the input represents a truly unfamiliar and novel situation so as to Bayesian methods in

Hey, I'm a mathematician, how can I not love it's right. But the beauty of Bayes is its simplicity. You're just right. The probability of two events A and B in two different ways. The probability of being given eight times the probability then the probability of a given p times the probability of B set them equal because they're identical. Divide by, let's say, probability A and then optimized. That's Bayes and it's a useful statistical method and should continue to be used in statistics. But it's just the form of identity where in lies its power. It says nothing about any physical reality, whether in physics, chemistry or biology. Today's rule itself tells us nothing about physical reality and contains no risk takes to discover anything about physical reality that you need to develop models driven by a profound analysis of large databases. So it turns out that biological models like R do not incorporate the base rule. However, authors routinely choose the best or optimal categories that represent the data best, so you don't need Bayes to achieve optimality. Also, Bayes works best in a stationary world with stationary probabilities and it's designed to learn about a non stationary one discusses till the cows come home. It's good for what it was designed for and some of the neuroscientists who try to apply Bayes wonderful experimentalists, but they know no math and no theory. And you know, it's the temptation of a free lunch. There it is waiting to be applied. There is no free lunch in science. Now, as a final point of comparison, what are the is Smith? I didn't replied. Yes, yes. Oh, okay. You quoted a sentence of Smith. But before that said Smith wrote, It is also worth highlighting that as our is intended primarily as a proof of concept and a demonstration of an available model expansion reduction approach that can be used within active inference research. It does not explicitly incorporate some aspects such as top down intention that are of clear importance to the cognitive learning processes and that have been implement in previous models. For example, the adaptive resonance. The Iraq model of Grossberg was designed to incorporate top down attentional mechanisms and feedback mechanism to address a fundamental knowledge acquisition problem, the temporal instability of previously learned information that can occur when a system also remains sufficiently plastic to learn new and potentially overlapping information. Well, our simulations do not explicitly incorporate these additional complexities. There are clear analogs. The top down and bottom up feedback exchanges are not within our model, such as the prediction and prediction error signaling within the neural process theory associated with active instrumental order addresses the temporal instable problem, primarily the mechanism that top down expectancies that guide attention and match them on platform up into patterns, which is quite similar to the prior expectations and likelihood mappings used with an active inference. But as I've already noted,

the quote prior expectations and likelihood matching mappings within adaptive inference, quote unquote do not have any of the key learning, attention and memory stability properties. The matching New York matching rule is a unique solution to that problem in its variations, it's been supported by psychological, anatomical, physiological and biophysical data. It also occurs in many species. Nobu Suruga, for example, shows it occurs in bats. It occurs in ferrets, you know. So also, I think it's important to note that when learning begins in an art model, it doesn't need for higher expectations or likelihood. In fact, typically the initial bottom up, it shows it to be random if you don't know what you're going to be experiencing. And the initial top down expectations are chosen to be large so that whatever category happens to be learned when it reads out its top down expectation, it can match whatever features activated that category. So they will start lodging than their room to match the critical features that happen to be learned in that category. So there are no built in models or discovers its own models. I should also emphasize that active inference is also not explainable. Sports is explainable because a currently active feature activity pattern, namely the features to which attention is paid, controls all learning and prediction. The model and in principle can be measured by neurophysiological experiments, a model without stellar activities or short term memory traces that can represent the critical feature and can't be explainable. So I think there are quality differences. I don't say people shouldn't use active inference. It may be incredibly useful and powerful in technological applications, but when one is doing, you know, brain science, psychology, it just doesn't match the foundational data. It's just doesn't want a personal thing. Thanks. And I guess you somehow already answered part of this question, but what are the possible ways in which art's approach to explainable AI, which, if I'm not mistaken, can be described as a modern dependent, intrinsically explainable approach, can inform active inference inferences approach across fertilize, but that which is based on objective reasoning through constructing generative models. as I sketched out in an pezzullo's understanding explanation and active inference paper. Well, first, I don't think or is model dependent. As I just noted, one begins typically to learn in art with random initial bottom up waves and uniformly distribute you did top down initial adaptive way so you can match any category that you happen to learn. But the authors you quoted write in part that active inference in here, I want to quote them so we can find a more detail. Active inference quote implies a deep generative model that includes a model of the world used to infer policy in a higher level model that attempts to predict which policies will be selected based upon a space of hyper that is counterfactual explanations and which can subsequently be used to provide retrospective

explanations about the policies pursued. So again, artworks without a generative model of the world or any predefined paths, the course one is trying to discover what changing world it happens to be, and then nobody knows what it is at priority and in general or not. Classify responds to a front end of three processes that process perceptual data from one or another sense, notably vision and audition, where we get most of our information. And that's why a classifier like art begins its work in the brain, in the temporal cortex, where it receives highly pre processed perceptual representation. So for decades, the work went into understanding. Brains consciously stay in here and in the case of vision or classifies perceptual boundaries and surfaces that require multiple stages of processing. Because as I mentioned briefly, they were the outcome of what I call hierarchical resolution of uncertainty. You need multiple stages to define a perceptual boundary surface one reason being because the sensory organs registers such noisy and incomplete data like you may know, there are photosensitive has a huge blind spot where you can't register any sense, any virtual signal. The blind as big as the phobia or all of our high resolution vision occurs. So it's not a little thing. And moreover, veins come out of the 12 year and occlude the retina in multiple places. And you can register visual signals on the veins, either So the signal you're getting is very incomplete and it takes multiple processing stages to overcome those uncertainties. And my cousin, I've worked for decades to explain how that happens and maybe I'll stop after so much. Now, this next question is of a personal interest to me, because currently working on modeling some probabilistic aspects of affective response to music and your most recent paper toward understanding the brain dynamics of music immensely helped me gain a better understanding of entrainment. As you pointed out in the supplementary notes for this paper, violation of prior learned expectations is instrumental in inducing a wide range of affective responses in musical and nonmusical situations. Some psychologists, such as Patrick Houston, have distinguished between perception and arousal of emotions. Trust you, but you left out a question. This is lots of time where you just skip the next question. I want up quite a bit about it so how do you see the future of art and neuro inspired? I think that will be our last question. Okay. So come back to that. Yes, yes. Thank you. Because that that's an important question. Yes. So sorry to interrupt. I just want to be sure no problem. Thank you. Now, yes. Well, some psychologists such as Patrick Forslund, have distinguish between the perception and the arousal of emotions in the context of musical experience. And also several studies such as The Works of Use, Lynn and Gabrielson from the Psychology Department of Sully University have shown that despite music's ability to communicate a wide range of positively and negatively emotions, it is

somehow evokes mostly positively balanced emotions. For instance, we can easily perceive rage or anger in music without necessarily getting angry. On the other hand, we're more likely to actually feel elevated and happy after listening to music. And also the evidence shows that this disparity between perception and evocation of emotion is probably even more significant in musical experiences than any other nonmusical experiences. So how can this difference in diversity between perceived and aroused, aroused or evoked musical emotions be accounted to be accounted for within art framework? Can it possibly be regarded as another kind of broken symmetry, as you mention on page 621 of your book, but specific to musical? So as you've noted, I haven't studied this issues in the context of music, but I'll try to venture some general comments I should note that the lab and art model, which is a development of art, to show how and why all neocortical circuits that support perception and cognition typically share a canonical six layer circuit. My colleagues and I have modeled, just as in our brains, variations of this canonical lamina circuit can support all perceptual and cognitive processes. So there's a major generalization of art, and we've done it for vision, speech and cognitive working memory and planning in particular. So the main point is that the alignment of circuitry is basically in all perceptual areas, vision or that's how one can create a context for discussing music. And in fact, my work on music applied such discoveries. I was able to put together discoveries that had been made based on what I believe were different evolutionary pressures that had the organization about brains. But that evolution also discovered and I try to sketch how if you put some them together in a certain way, then the capacity for and consciously music could arise. So now how about arousal? So it's essential, of course, to all awareness and consciousness. Your cortex needs to be adequately aroused for waking consciousness to occur at all. It also arousal plays a major role in the processing of emotions and is very relevant. The musical issues was mitigated dipole model explains how own and plus this piece of this that's organized in all parts of our brain perceptual cognitive motor affective in particular, emotions are organized in pairs as in such an emotional dipole. And one reason is because emotions need to compete with each other, such as fear versus relief. For example, in post-traumatic stress disorder therapy, a therapist may try to help patients to think about positive experiences that generate relief in order to inhibit the chronic that's so destabilizing during PTSD. So their opposites are competing and. Another property that arousal enables is that this sudden offset of an emotion like here, let's say during escape behavior, let's say, you know, favorite example of mine is, you know, some cruel experimentalist puts the picture in a box that was electrified. The pitch is feeling pain and fear. It's dashing frantically around, trying to keep its feet off the

floor. It bangs into a red buzzer, the buzzer shuts the shock off and the animal experienced a wave of relief or positive motivation for learning the escape response. So this rebound from theater relief that can be associated with actions that lead to escape and can motivate escape by means in the future is energized by arousal. Indicate a dipole. You shut off the external cue of shock, but the arousal is technically or has sustained activity in both this gear and the channel. So because the arousal is sustained or tonic through time and equally activates the fear and relief channels when fear suddenly decreases, then arousal and the relief channel wins, the competition. And therefore I call what I call an antagonistic rebound from fear relief that activates the relief channel and thereby providing motivation for escape, whether reactive or learned to escape experiences. And I want to appropriate, which is related to some degree the music that the non occurrence of an expected event and by itself cause a burst of arousal and that's an antagonistic rebound and can slap emotions from the negative in so doing. And I always love that discovery because I especially loved discoveries where the occurrence of nothing has profound effects on future behavior. So it's the non occurrence, the expectation because of this mismatch that can emotions now how this influences the perception of music needs more work. It's as I mentioned my paper I haven't tried to study the I my my paper on music I feel is like you know a drop in the bucket. And hopefully if I don't get around to it someone else will. But the above examples show that arousal and emotion are not the same thing because arousal can support all emotion, peer relief, hunger, satiety, whatever the ones that when the competition or then able to support compatible behaviors by motivating them. And I've also explained that the level of arousal must be chosen within an intermediate range to support normal behaviors. It's the kind of golden mean. There's an inverted U on the effects of having arousal from too little or too much. And if you have too little arousal, you have an under arouse syndrome which can support symptoms like autism. And although arousal can support symptoms of abuse like schizophrenia, there's only one of many factors in these diseases. But I'm happy to say that subsequent clinical data supported those predictions that these two mental disorders are the opposite speeds of the arousal in gratitude. So I think and you know, this is very speculative because I've never really seriously studied it. And I try not to speculate, but the kind of arousal that music activates are generally positive, just like the arousal that activates exploratory behaviors is positive, is somehow linked to a a sonic adventure, if you like. There's no aversive cue when listening to music, except perhaps music played so loud as to cause a headache or your damage or even a seizure. In susceptible individuals, though there's no particular reason why it shouldn't be positive. So now which

question you want to ask me? I think we're left with just two other questions. If you're not too tired to start with. As a final point, a person noticed that on a more philosophical. Yes, the last where until last the two questions. So you want to uh how do you see the future of our neurons but our research. And before that. Yes, before that I just wanted to ask you about the your view about artificial consciousness. Do you see. I really need the first answer. The answer the second question. Okay. As you wish. Yes. So so. Yeah. And how do you see. Yeah, yes, it's quite fine. And how do you see the future of art and brain inspired A.I. research in general in view, what research areas ought to gain more attention than they do today? So I'll give you quite a general answer, but it implies what I think about this. So first, with a caveat, I like to say I couldn't predict the present, so I can't predict the future. That being said, I believe that all engineering technology and I will increasingly embody autonomous adaptive intelligence in the coming century, and we can already see its beginnings in autonomous automobiles and and increasingly autonomous controllers on the factory floor. And many people have written about it. And I think art will play a central role in this, as well as other models that are summarized in my book. And that's because already in 1980, I published a thought experiment in the journal Psychological Review, which was then and still probably remains the leading theory journal in psychology. And you may recall that Einstein derived both special relativity theory and general relativity from Ford experiments. And let me just clarify that my thought experiment, when they describe their enormous power. So my thought experiment was about how any system an autonomous fleet what about autonomous correct predictive errors in a changing world and the hypotheses upon which what experiment were derived. But just a few facts that are familiar to us all from daily life. And they are familiar because represent ubiquitous environmental pressures on the evolution of our brains over the millennia. And when they act together, I think just art is the unique outcome. That's a huge claim and I turn to the power of the thought experiment. Not any personal ego trip without belief in particular, nowhere in the thought experiment of the word mind or brain mentioned. So if you accept that these facts about the world exist, which we all do, and that they're always operating on us, then you have to accept the outcome. If you believe in the scientific method logic. So what is a universal solution to the problem of economists? Error correction a change in world turn another way If you can find a mistake in the thought experiment, then I think you either have to believe in art, like dynamics may be expressed in life, in art or your favorite art variant. Or you have to give up your belief in logic and the scientific method. It doesn't imply that I can't be further developed. I expect a large number of scientists to acknowledge and technologies to

be busy developing, much like architecture's long instrument. So maybe now we can go to your philosophical note with that. Yes. Thank you so much. Now, just as I mentioned, I mentioned earlier, I just wanted to ask about your view about your view on artificial consciousness. Do you see the consciousness as artificially reducible or engineer able as some researchers like Mike Soames leads? Is there a fundamental distinction between biologically conscious agent and an artificial agent with a fully simulated computational model of consciousness? I know it's a big, big question, but I couldn't resist asking your opinion as an authority on consciousness modeling. I'm happy to give it a shot. So as I just noted, my work on art suggests itself the universal about how we can learn to correct predictive errors in a changing world. My work also shows an analysis of hierarchical resolution. Remember, like how you go from sun, a noisy retina to the surface representation that can control looking in reaching how evolution may have been driven to discover conscious states. So this was a surprise to me too. Conscious states were needed in order to choose that processing level. The levels that computes the sufficiently complete context sensitive and stable representation in the case of vision, the surface representation with which to successfully plan and act to value calls. So let me make it clear. So you start with a noisy rat and you have to go up all of these stages until you get sufficiently complete surface and boundary representation that you can use to regulate successful action. And if you used one of the stages, it would lead to incorrect actions which would kill you off by Darwinian selection. So how the hell do you know where the stage is, where you can compute this sufficiently complete, extensive and stable one and I proposed in vision, I predicted that the choice is embodied in what I call surface shroud resonance between destroyed visual cortical area, the floor and the next processing stage. A stirrup while cortex of the state. So it's in the poor. You get this really good surface representation and then a resident between the surface and spatial attention which fits the surface, that spatial attention and PPC is called a shroud of the pilot gave it that name. A surface shroud resonance allows you to a conscious spatial attention to the surface that you're going to use the controller looking and reaching behaviors. So it's a way of ensuring you have a good enough representation to control action. So the shroud is computed in plastic parietal cortex, which is part of the dorsal or where cortical stream and the shroud modulates invariant category learning in the ventral or what cortical stream? I can't go into that now, but my book discusses it. The category of learning itself in the what cortical, as I indicated, is supported by a feature category resonance. And so the surface shroud resonance is modulating invariant category learning in the featured category resonance with surface shroud resonance also

supports in the future category resonance is supporting conscious recognition and when they synchronize across streams on a familiar object, that's when you consciously see something that you know about. Okay, so conscious states here arise due to learning requirements. This sort of fell out of the wash of how you do invariant category learning and learning in particular. It's regulating feature category resonance. So given that the above solutions the computational universe so in the sense I sketch the self-organizing machine that embodies them should be able to support internal representations with parametric properties mimic the state. So whether such a machine can experience conscious qualia remains as much of a mystery to machines as it does for humans. And that's because no computational theory, which after all, is just a set of equations, can't do more than imitate the dynamics of our brains. Perhaps with great precision. I don't have a clue why the representations that I've worked so hard to explain huge amounts of psychophysical data about them excruciating 3D form. Just go through the list why they support Qualia. I don't know. Ask God, or whatever God you choose to believe in in the 21st century. Thank you so much, Professor. I think we have a couple of questions in the chat but we are actually approaching our two hour limit. I don't know, Daniel, if it's a good place to stop or whatever you say. I think that's great place to stop. You've given us a lot to think about and digest. And I hope that these words are taken well and paid attention to might create some categories, activate some categories. But Professor Grossberg, thanks again for this amazing livestream. We really appreciate it. Appreciate it. And I'm depending on younger people like yourself to do just what you said, Daniel! I'm not going to be around that much longer. So I hope you have, whether with my work directly or related work, you have a very fulfilling intellectual adventure. I know I've been on a wild ride since I was 17, and that's 65 years of discovery.